Feynman Diagrams and Scaling Laws for Low-Rank Multilayer Neural Networks

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Abstract

We develop a finite-width Feynman theory for multilayer networks in which a frozen left factor is composed with a trainable right factor. This models the isometrically whitened effective matrices between consecutive multi-component and multi-layer neural-network (MMNN) blocks. Orthogonal Weingarten calculus turns the frozen factor into connected cumulant vertices of the normalized Haar projector $G = (n/r)UU^\top$, whose exact covariance has defect scale $\gamma_{n,r} = n(n-r)/[r(n-1)(n+2)]$. The calculus has an exact dense endpoint: when $r = n$, every centered projector vertex vanishes and the empirical NTK agrees pathwise with the dense model. At ReLU criticality we derive the tensor hierarchy $V_4 = O(\bar{\varepsilon}L)$, $(D, F) = O(\bar{\varepsilon}L^2)$, and $(A, B) = O(\bar{\varepsilon}L^3)$, and reconcile it with all thirty componentwise effective exponents reported in the finite-width NTK experiments of Guillen, Misof, and Gerken. Matrix concentration then gives the NTK perturbation scale $s_{L,n,r} \lesssim L^{3/2}\sqrt{m\bar{\varepsilon}} + mL^3\bar{\varepsilon}$, up to logarithms.

For data covariance eigenvalues $\eta_j \asymp j^{-(1+\alpha)}$, we combine this bound with Weyl and Davis–Kahan theory. Only the near-degenerate tail can become Wigner-like: the stable head ends at $J_{\mathrm{mix}} \asymp (\eta_1/s_{L,n,r})^{1/(2+\alpha)}$. We compute the random-tail contribution to the entire learning curve exactly: a block-Haar eigenbasis has closed-form risk mean and variance from the orthogonal fourth moment. This yields a mixing time $t_{\mathrm{mix}} \asymp P^{-1}s_{L,n,r}^{-(1+\alpha)/(2+\alpha)}$ and a sufficient rank law protecting the aligned early-stopping optimum, $r \gtrsim mL^3(P\beta)^{2(2+\alpha)/(1+\alpha)}$, up to constants and logarithms. Simulations verify the exact Weingarten moments, spectral mixing threshold, data-scaling bend, and early-stopping shift. Proofs are organized after the diagrammatic derivation so that the physical expansion and its rigorous probabilistic justification remain separately auditable.

1 Introduction

Low-rank neural networks are usually motivated by efficiency: fewer parameters, lower memory traffic, and cheaper adaptation. This motivation is incomplete. Rank also changes the geometry of training. In the NTK regime, gradient descent is close to kernel regression when the empirical NTK Gram matrix is positive and stable. Thus a low-rank network is useful not only when it is small, but when its finite NTK stays close enough to a positive limiting kernel. We call this the edge of convexity: the regime where the network is finite and structured, but the NTK Gram matrix still provides a convex local training geometry.

This question is difficult because three limits interact. Width controls ordinary finite-width fluctuations, depth amplifies NTK tensor corrections, and rank changes the channel contractions themselves. Treating low rank as merely replacing $1/n$ by $1/r$ misses the full-rank endpoint: if a rank-n factorization is just an orthogonal change of variables, the low-rank correction must vanish exactly. Conversely, bottleneck random-feature low-rank networks do not reduce to dense MLPs and can suppress old NTK paths geometrically. A useful theory must separate these two cases.

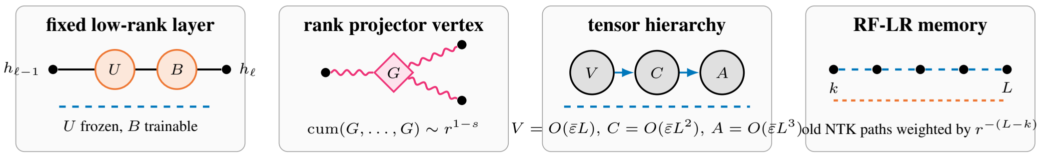

Figure 1: Main diagrammatic objects, drawn in the color and line vocabulary used throughout the finite-width NTK diagram literature. Solid black lines denote NNGP/preactivation transport, dashed colored lines denote NTK legs, orange and pink vertices denote rank and Haar-projector cumulants, and gray blobs denote transported tensor cumulants.

We study scalar-output ReLU networks with layer weights of the form

$$W^{(\ell)} = \frac{\sigma_w}{\sqrt{n_{\ell-1}}}\sqrt{\frac{n_\ell}{r_\ell}}\,U^{(\ell)}B^{(\ell)}, \qquad (U^{(\ell)})^\top U^{(\ell)} = I_{r_\ell}, \tag{1}$$

where $U^{(\ell)}$ is frozen Haar-Stiefel and $B^{(\ell)}$ is trainable. Haar-Stiefel means that the columns of $U^{(\ell)}$ form an orthonormal r_ℓ-frame in $\mathbb{R}^{n_\ell}$, chosen uniformly among all such frames. Equivalently, $U^{(\ell)}$ selects a random r_ℓ-dimensional subspace with no preferred direction. When $r_\ell = n_\ell$, the map $B^{(\ell)} \mapsto U^{(\ell)}B^{(\ell)}$ is an orthogonal reparametrization of a dense MLP. Thus the full-rank endpoint is exact, not asymptotic. Away from full rank, the relevant random object is the normalized projector

$$G = \frac{n}{r}UU^\top, \tag{2}$$

whose covariance scale is

$$\gamma_{n,r} = \frac{n(n-r)}{r(n-1)(n+2)} = \frac{1}{r} - \frac{1}{n} + O(n^{-2}). \tag{3}$$

The mixed perturbation parameter is therefore $1/n + \gamma_{n,r}$. The term $1/n$ is the ordinary dense finite-width correction, while $\gamma_{n,r}$ is the rank defect. The latter vanishes at $r = n$, as required by exact full-rank compatibility.

The connection to MMNNs is algebraic. An MMNN block is

$$h_\ell(x) = A_\ell\sigma(W_\ell h_{\ell-1}(x) + b_\ell), \qquad W_\ell, b_\ell \text{ frozen}, \qquad A_\ell \text{ trainable}, \tag{4}$$

with component dimension $r_\ell \ll n_\ell$ [13]. Between two nonlinearities the effective matrix is $W_{\ell+1}A_\ell$: a frozen left factor followed by a trainable right factor. Hence a depth-L MMNN is a concatenation of the fixed-left/trainable-right two-layer objects for which our rank-state Feynman calculus closes. A QR decomposition of Gaussian $W_{\ell+1}$ separates its radial Wishart part from a Haar-Stiefel orientation; equation (1) is exact for the whitened orientation, or equivalently after using the induced metric on A_ℓ. An unwhitened MMNN has additional fixed radial/Wishart vertices and is not pathwise identical to the Stiefel model. A strict nonnegative factorization changes the tangent cone and is not claimed here.

This MLP-compatible model should not be confused with a random-feature low-rank bottleneck (RF-LR). In that regime, the limiting NTK recursion contains an explicit factor $1/r$ on old NTK lines. At ReLU edge of chaos, old memory from layer k to layer L is suppressed as $(2r)^{-(L-k)}$. This creates a different spectral design rule: low rank shortens memory, but the centered deterministic gap is also smaller. The paper is organized around this separation.

Contributions.

- We prove exact full-rank matching: at $r = n$, the factorized network and dense MLP have identical empirical NTKs pathwise.
- We identify the low-rank Feynman vertices as connected cumulants of $G = (n/r)UU^\top$, with second cumulant scale $\gamma_{n,r}$.
- Under explicit ReLU transport assumptions, we derive tensor scalings $V = O(\bar{\varepsilon}L)$, $C = O(\bar{\varepsilon}L^2)$, and $A = O(\bar{\varepsilon}L^3)$, with endpoint constants 5, 21/20, and 173/720 for isolated off-diagonal modes.

- Under an explicit well-spread data assumption, we prove operator concentration for finite empirical NTKs at scale $L^{3/2}\sqrt{m\bar{\varepsilon}} + mL^3\bar{\varepsilon}$, up to logarithmic factors, and show that dNTK/ddNTK descendants introduce no new random vertices.
- We derive a conservative rank-depth stability rule for bottleneck low-rank dynamics: to preserve the centered NTK spectral margin in the unprojected RF-LR model, rank scales cubically with depth, $r \gtrsim mL^3$, up to logarithmic factors. We also state the separate centered/angularized mL^2 rule with its additional projection hypothesis.
- For power-law data spectra, we prove a stable-head/random-tail transition, compute the block-Haar tail learning curve exactly by orthogonal Weingarten calculus, and derive the early-stopping protection law $r \gtrsim mL^3(P\beta)^{2(2+\alpha)/(1+\alpha)}$.
- We validate the theory with proxy experiments, true finite-network autodiff NTKs, exact Haar moment tests, spectral eigenvector mixing, data-scaling curves, and early-stopping shifts.

Status of the claims. The paper separates what is proved algebraically from what is conditional on standard concentration hypotheses. Full-rank matching, the fixed-left gradient decomposition, the Haar covariance, and the scalar endpoint moments are unconditional calculations. The depth hierarchy and operator bounds are stated under the ReLU edge-of-chaos transport and well-spread data assumptions made explicit below. Direct finite-network cumulants are treated as diagnostics of the polynomial hierarchy, not as direct measurements of isolated Feynman coefficients.

2 Related Work

The NTK was introduced as the infinite-width kernel governing gradient-flow training [1]. This line of work changed the community's view of overparameterized networks: in the infinite-width lazy regime, training can be studied through a deterministic kernel, and convergence depends on the spectrum of the NTK Gram matrix. Follow-up work showed that wide networks of any depth evolve approximately as linear models around initialization [2], and that overparameterized networks can be trained globally under width conditions that keep the empirical NTK stable [3, 4, 5]. These results explain why positive NTK spectra are powerful, but they mostly treat dense networks and do not describe the finite rank-width corrections introduced by factorized layers.

A second thread studies the limits and finite-width corrections of the NTK. Tensor programs give a general language for infinite-width limits and parametrization choices across architectures [8, 9]. The neural tangent hierarchy and related expansions describe how finite-width dynamics depart from the frozen-kernel limit [10]. Feynman-diagram methods give a complementary perturbative calculus for wide networks, organizing corrections by connected cumulants and tensor contractions [11, 12]. Our work follows this finite-width viewpoint, but replaces dense-channel contractions by cumulants of the Haar projector induced by low rank.

MMNNs compose shallow random-feature maps whose inner weights are fixed and whose component-mixing matrices are trained [13]. Their adjacent product $W_{\ell+1}A_\ell$ is the motivating fixed-left/trainable-right low-rank matrix. Benigni and Paquette characterize the limiting NTK eigenvalue distribution of a two-layer network in a quadratic sample–width regime using free multiplicative convolution [14]; our object is complementary, because we concatenate such low-rank blocks and retain finite-width connected corrections. For generalization, Kramp, Lindner, and Helias derive power-law learning and early-stopping laws by dynamical mean-field theory [15]. We quantify when the finite-rank Feynman perturbation preserves those laws and when near-degenerate spectral eigenvectors randomize.

The closest optimization and spectral guarantees are complementary rather than substitutive. Prior work proves global convergence for deep networks with one wide layer followed by a pyramidal topology, showing that not every layer must be wide [6]. Tight lower bounds are also known for the smallest eigenvalue of deep ReLU NTKs, including finite-width architectures with one layer of order $\widetilde{\Omega}(m)$ and other widths essentially arbitrary [7]. These results already cover strong "one wide layer," bottleneck-width, global-convergence, and $\lambda_{\min}$-type guarantees for full-rank networks. Our claim is different: we do not try to improve their global convergence conditions. We provide a finite-rank perturbation theory for factorized layers $W = \sqrt{n/r}\,UB$, isolate the exact rank-defect parameter $\gamma_{n,r}$, prove exact full-rank matching at $r = n$, and convert the finite-rank NTK deviation into a Weyl spectral-preservation criterion. Recent quantitative GP approximation during training [20] is also orthogonal: it controls Wasserstein distance to a Gaussian-process limit for trained shallow networks, whereas our object is operator concentration of $\widehat{\Theta} - K_*$ for deep low-rank NTKs and the resulting stability of the centered spectral margin.

A third thread concerns practical NTK computation and low-rank structure. Empirical NTKs can be computed efficiently enough to probe finite networks directly [18]. Low-rank adaptation and low-rank training have also been analyzed in NTK-like regimes, including results showing that sufficiently large low-rank adapters avoid bad local minima in certain settings [19]. These works motivate low rank as an optimization and adaptation tool. Our contribution is different: we ask when low rank preserves the NTK convexity certificate at finite width and finite depth, and we show that the answer depends on whether the rank constraint is an isometric perturbation of the dense MLP or a genuine RF-LR bottleneck.

3 Models and Full-Rank Matching

Let $x_1, \ldots, x_m$ be a fixed dataset. The empirical scalar-output NTK is

$$\widehat{\Theta}_{ab} = \nabla_\theta f(x_a) \cdot \nabla_\theta f(x_b). \tag{5}$$

For the factorized model (1), trainable parameters are the right factors $B^{(\ell)}$ and the readout. We deliberately keep the left Stiefel factors $U^{(\ell)}$ frozen. This is not only a computational choice. It is what makes the low-rank model a controlled NTK perturbation: the trainable tangent space is a fixed linear subspace, and all rank-dependent randomness enters through the static projector $G = (n/r)UU^\top$. If U is trained as well, the tangent space itself moves, producing additional dNTK terms that couple variations of U and B.

Theorem 3.1 (Pathwise full-rank matching)**.** *If $r_\ell = n_\ell$ for every hidden layer, then for every realization, every dataset, and every depth, the empirical NTK in B-coordinates equals the empirical NTK of the dense MLP in dense-weight coordinates $\widetilde{W}^{(\ell)} = U^{(\ell)}B^{(\ell)}$. The same equality holds for the NNGP, dNTK, ddNTK, and all tensor cumulants built from them.*

Proof. When $r = n$, U is orthogonal. The linear map $\Delta B \mapsto U\Delta B$ preserves Frobenius inner products. The Jacobian with respect to B is the dense-weight Jacobian composed with this isometry; hence $J_B J_B^\top = J_W J_W^\top$. Higher derivatives and cumulants are invariant under the same coordinate change. ☐

Proposition 3.2 (Why the left factor is frozen)**.** *Consider one layer $W = \alpha UB$, with $U^\top U = I_r$, $\alpha > 0$, and loss gradient $H = \partial\mathcal{L}/\partial W$. If only B is trained, then*

$$\nabla_B \mathcal{L} = \alpha U^\top H, \qquad \dot{W}_B = -\eta\alpha^2 UU^\top H. \tag{6}$$

The layer motion is the fixed projected gradient through $UU^\top$. If U is also trained on the Stiefel manifold, then the projected Riemannian gradient contributes

$$\dot{W}_U = -\eta\alpha^2(I - UU^\top)HB^\top B \quad \textit{up to rotations inside } \operatorname{span}(U) \textit{ that can be absorbed into } B. \tag{7}$$

Consequently the empirical NTK and its time derivative contain additional U-B interaction blocks, proportional to contractions with $B^\top B$. These blocks are absent in the frozen-left model and are not controlled solely by Haar-projector cumulants of $G = (n/r)UU^\top$.

Proof idea. With U fixed, the only admissible layer variation is $dW = \alpha U\, dB$, so gradient flow in B moves the dense weight through the fixed projector $UU^\top$. If U is trained, the Stiefel constraint allows normal motions of the subspace in directions $(I - UU^\top)$, and these motions are multiplied by $B^\top B$ after mapping back to $W = \alpha UB$. Thus the tangent kernel is no longer a fixed-projector B-kernel: it contains moving-subspace and U/B cross blocks. The full differential calculation is given in Appendix A. ☐

Proposition 3.2 explains the modeling choice. Training both factors may be useful in practice, but it is a different dynamical system: the low-rank subspace rotates during training, and the NTK drift receives extra moving-projector terms. Freezing U isolates the rank-width effect we can prove and keeps the diagrammatics closed under the Haar projector vertices of Lemma 4.1.

The RF-LR bottleneck has a different recursion,

$$\Theta^{(\ell)} = 1 + \frac{1}{r}\Theta^{(\ell-1)}\dot{\Sigma}^{(\ell)} + \frac{1}{r}\Sigma^{(\ell)}. \tag{8}$$

At the ReLU fixed point $\dot{\Sigma}^{(\ell)} \to 1/2$, so an old NTK contribution is multiplied by approximately $(2r)^{-(L-k)}$. This is a different model, not a perturbation of the dense MLP.

4 Haar Projector Vertices

Let $U \in \mathbb{R}^{n\times r}$ be Haar-Stiefel and $G = (n/r)UU^\top$. Orthogonal invariance gives the full covariance tensor.

Lemma 4.1 (Haar projector covariance)**.** *For all indices,*

$$\mathrm{Cov}(G_{ij}, G_{kl}) = \gamma_{n,r}\left(\delta_{ik}\delta_{jl} + \delta_{il}\delta_{jk} - \frac{2}{n}\delta_{ij}\delta_{kl}\right), \qquad \gamma_{n,r} = \frac{n(n-r)}{r(n-1)(n+2)}. \tag{9}$$

In particular $\gamma_{n,n} = 0$.

Proof. The covariance of a Haar projector has the invariant form $a(\delta_{ik}\delta_{jl} + \delta_{il}\delta_{jk}) + b\delta_{ij}\delta_{kl}$. The identity $\operatorname{tr} G = n$ gives $2a + nb = 0$. The standard second moment of a Haar projector gives $a = \gamma_{n,r}$, hence (9). □

Higher cumulants are given by orthogonal Weingarten contractions. Connected s-point cumulants scale as $O_s(r^{1-s})$ and vanish in the full-rank endpoint. Thus the low-rank Feynman rule is simple: dense Gaussian-ReLU vertices remain, and each low-rank correction inserts connected cumulants of G.

Lemma 4.2 (Rank cumulant power counting)**.** *Fix a layer ℓ and condition on all previous layers. For rank channel a, let*

$$Z_a^{(\ell)} := \left\{ \left(g_a^{(\ell)}(x_u), \sigma(g_a^{(\ell)}(x_u)), \sigma'(g_a^{(\ell)}(x_u))\right)_{u=1}^m, \text{ and the corresponding derivative atoms} \right\}.$$

Here $g_a^{(\ell)}(x_u)$ is the scalar preactivation of rank channel a on sample x_u, after the deterministic conditioning on layer $\ell - 1$. Thus $Z_a^{(\ell)}$ is the full local random object from which one-channel quantities such as $\sigma\sigma$, $\sigma\sigma'$, $\sigma'\sigma'$, and derivative insertions are evaluated. The variables $Z_1^{(\ell)}, \ldots, Z_r^{(\ell)}$ are conditionally iid. To simplify notation, write them as Z_a, and assume the local functions below have uniformly bounded moments of all orders. Define empirical observables

$$X_i^{(r)} = r^{-a_i}\sum_{a=1}^{r} f_i(Z_a), \qquad i = 1, \ldots, s.$$

Then the connected cumulant satisfies

$$\mathrm{cum}\left(X_1^{(r)}, \ldots, X_s^{(r)}\right) = r^{1-\sum_i a_i}\ \mathrm{cum}\left(f_1(Z), \ldots, f_s(Z)\right). \tag{10}$$

In particular, normalized empirical averages have s-point cumulants of order r^{1-s}.

Proof. By multilinearity of cumulants,

$$\mathrm{cum}(X_1^{(r)}, \ldots, X_s^{(r)}) = r^{-\sum_i a_i}\sum_{a_1,\ldots,a_s=1}^{r} \mathrm{cum}(f_1(Z_{a_1}), \ldots, f_s(Z_{a_s})).$$

Independence kills every term whose indices are not all in the same connected block. Only $a_1 = \cdots = a_s$ contributes, giving r identical terms and hence (10). □

This elementary lemma is the right-factor-only analogue of the usual width power counting. It explains why finite-rank bias terms are typically $O(r^{-1})$ after averaging over seeds, while single-seed fluctuations are $O_{\mathbb{P}}(r^{-1/2})$. It also explains why the exact full-rank endpoint must be treated with the projector defect $G - I$, rather than by the informal replacement $n \mapsto r$.

5 Tensor Scaling at ReLU Edge of Chaos

For normalized ReLU, the edge-of-chaos correlation asymptotics used in the RF-LR analysis [17] give $\theta_\ell = 3\pi/\ell + O(\log \ell/\ell^2)$, and

$$\dot{\mathcal{C}}(\rho_\ell) = 1 - \frac{3}{\ell} + O\left(\frac{\log \ell}{\ell^2}\right). \tag{11}$$

Transporting an external NTK leg from layer k to L gives the factor $(k/L)^\lambda$ for an exponent $\lambda \in \{0, 3\}$, depending on whether the Gram entry is diagonal or off-diagonal.

Lemma 5.1 (Depth-transport accumulation). *Let $T_\ell \in \mathbb{R}^d$ satisfy*

$$T_{\ell+1} = \left(I - \frac{M}{\ell} + E_\ell\right) T_\ell + \ell^\beta (s + e_\ell), \tag{12}$$

where $\|E_\ell\| \le C \log^p(\ell)/\ell^2$, $\|e_\ell\| \le C \log^q(\ell)/\ell$, and $M + (\beta+1)I$ is invertible. Then

$$T_L = L^{\beta+1} \left(M + (\beta+1)I\right)^{-1} s + O\left(L^\beta \log^{p+q+2} L\right). \tag{13}$$

Proof. Iterating (12) gives a variation-of-constants formula. The product of propagators from k to L is $(k/L)^M$ up to a multiplicative error whose accumulated contribution is $O(L^\beta \log^{p+q+2} L)$, since $\sum_k \log^p(k)/k^2 < \infty$. Thus

$$T_L = L^{-M} \sum_{k \le L} k^{M+\beta I} s + O(L^\beta \log^{p+q+2} L).$$

Euler-Maclaurin yields

$$L^{-M} \sum_{k \le L} k^{M+\beta I} = L^{\beta+1} \int_0^1 u^{M+\beta I} \, \mathrm{d}u + O(L^\beta),$$

and the matrix integral is $(M + (\beta+1)I)^{-1}$. □

Let $\bar{\varepsilon} = \max_{\ell \le L}(1/n_\ell + \gamma_{n_\ell, r_\ell})$. The finite-width tensors obey

$$V_L = O(\bar{\varepsilon} L), \qquad C_L = O(\bar{\varepsilon} L^2), \qquad A_L = O(\bar{\varepsilon} L^3), \tag{14}$$

where C denotes the mixed NNGP-NTK block (D, F), and A denotes the NTK-NTK block (A, B).

Proposition 5.2 (Conditional tensor hierarchy). *Assume the low-rank cumulant recursion closes on the same tensor sectors as the dense finite-width Feynman hierarchy and that the ReLU edge-of-chaos transport is uniform away from the diagonal singularity. Then the isolated tensor coordinates satisfy*

$$V_L = O(\bar{\varepsilon} L), \qquad D_L, F_L = O(\bar{\varepsilon} L^2), \qquad A_L, B_L = O(\bar{\varepsilon} L^3). \tag{15}$$

Consequently, the perturbative mean-NTK correction is controlled in the regime $L^3 \bar{\varepsilon} \ll 1$.

Proof. The fresh NNGP four-point source contributes $O(\bar{\varepsilon})$ per layer and has no transported NTK leg in the dominant radial mode, so summation gives $V_L = O(\bar{\varepsilon} L)$. The mixed blocks D, F have one transported NTK leg and therefore one depth-transport accumulation with $\beta = 1$, yielding $O(\bar{\varepsilon} L^2)$ by Lemma 5.1. The NTK-NTK blocks A, B have two transported NTK legs, equivalently source exponent $\beta = 2$, giving $O(\bar{\varepsilon} L^3)$. These are isolated tensor coordinates; scalar-output cumulants can mix these sectors. □

The exponents in the published plots. Table 1 records every critical ReLU power fitted in the vector figures of Guillen, Misof, and Gerken [12]. Their networks have depths only up to 30, and the regressions begin between layers 5 and 12, so these are effective slopes $p_{\text{eff}}(L) = \mathrm{d} \log |T_L| / \mathrm{d} \log L$, not new diagrammatic degrees. The thirty fits cluster by sector: V_4 around the asymptotic degree one, D, F around degree two, and A, B around degree three. Off-diagonal components approach later because the ReLU correlation satisfies $1 - \rho_L = O(L^{-2})$, while its derivative transport has $O(L^{-1} \log L)$ corrections. If $T_L = cL^p(1 + aL^{-\omega} + o(L^{-\omega}))$, then

$$p_{\text{eff}}(L) = p - \omega a L^{-\omega} + o(L^{-\omega}); \tag{16}$$

hence finite-depth fits above p are compatible with a negative first correction and do not alter the leading Feynman count.

Table 1: Componentwise critical-depth exponents read from the regression labels in the five vector plots of Guillen, Misof, and Gerken [12]. The last row is the leading degree predicted by the transport diagrams.

indices	V_4	D	F	A	B
0000	1.19	2.16	2.29	3.22	3.30
0101	1.31	2.41	2.74	3.47	3.78
0022	1.49	2.57	2.36	3.82	3.34
0103	1.42	2.39	2.70	3.45	3.72
0023	1.47	2.58	2.74	3.76	3.61
0123	1.55	2.53	2.78	3.62	3.81
diagrammatic degree	1	2	2	3	3

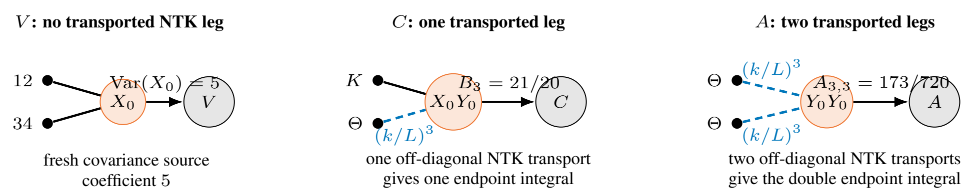

Figure 2: Endpoint diagrams for the constants in (19). A V-source has no transported NTK leg and gives $\mathrm{Var}(X_0) = 5$. A mixed C-diagram has one off-diagonal NTK leg transported by $(k/L)^3$, giving $B_3 = 21/20$. An A-diagram has two such transported legs, giving $A_{3,3} = 173/720$.

Endpoint constants. At the ReLU endpoint, set $X_0 = 2(g_+)^2$ and $Y_0 = 2\mathbf{1}_{\{g>0\}}$. Then

$$\mathrm{Var}(X_0) = 5, \qquad \mathrm{Cov}(X_0, Y_0) = 1, \qquad \mathrm{Var}(Y_0) = 1. \tag{17}$$

The three constants correspond to the following endpoint diagrams: For a transported off-diagonal leg, $\lambda = 3$. Indeed, the leg is multiplied from layer k to layer L by

$$\prod_{j=k+1}^{L} \dot{\mathcal{C}}(\rho_j) = \prod_{j=k+1}^{L} \left(1 - \frac{3}{j} + O\left(\frac{\log j}{j^2}\right)\right) = \left(\frac{k}{L}\right)^3 (1 + o(1)),$$

because the logarithm of the product is $-3\sum_{j=k+1}^{L} j^{-1} + O(1/k)$. This is the source of the exponent 3. The mixed and NTK-NTK endpoint integrals are

$$B_\lambda = \frac{1}{\lambda+1}\left(5 - \frac{4}{\lambda+2}\right), \qquad A_{\lambda,\mu} = \frac{1}{(\lambda+1)(\mu+1)}\left(5 - \frac{4}{\lambda+2} - \frac{4}{\mu+2} + \frac{4}{\lambda+\mu+3}\right). \tag{18}$$

Thus for off-diagonal modes,

$$V: 5, \qquad C: B_3 = \frac{21}{20}, \qquad A: A_{3,3} = \frac{173}{720}. \tag{19}$$

These constants are tensor coefficients after pairing-basis extraction, not arbitrary scalar-output cumulants. Direct finite-network cumulants mix Wick/Wishart terms, radial modes, readout contractions, and several Feynman sectors; they are therefore expected to show finite-depth effective slopes rather than the isolated constants exactly.

5.1 A three-layer finite-rank calculation

The simplest place where the low-rank simplification is visible is a three-layer network, i.e. two ReLU feature maps. Let

$$Q_x = \frac{1}{r}\sum_{a=1}^{r} X_a^2, \qquad Q_y = \frac{1}{r}\sum_{a=1}^{r} Y_a^2, \qquad S = \frac{1}{r}\sum_{a=1}^{r} X_a Y_a, \qquad C_r = \frac{S}{\sqrt{Q_x Q_y}}.$$

For the right-factor-only low-rank model, the empirical second hidden-layer kernel has the schematic form

$$K_{2,r} = \sqrt{Q_x Q_y}\,\kappa(C_r), \qquad \dot{K}_{2,r} = \dot{\kappa}(C_r), \qquad \widehat{\Theta}_r^{(3)} = K_{2,r} + S\,\dot{K}_{2,r},$$

where κ is the normalized ReLU covariance map.

Proposition 5.3 (Delta-method correction for three layers)**.** *For fixed correlation $\rho \in (-1, 1)$, there is an explicit smooth function $\Delta^{(3)}(\rho)$, depending only on $\kappa, \dot{\kappa}$ and the covariance matrix of (X^2, Y^2, XY), such that*

$$\mathbb{E}\,\widehat{\Theta}_r^{(3)}(\rho) = \Theta_\infty^{(3)}(\rho) + \frac{1}{r}\Delta^{(3)}(\rho) + O(r^{-2}), \qquad \widehat{\Theta}_r^{(3)} - \mathbb{E}\widehat{\Theta}_r^{(3)} = O_{\mathbb{P}}(r^{-1/2}). \tag{20}$$

For scale-invariant ReLU on the diagonal, $\Delta^{(3)}(1) = 0$.

Proof. Write $\widehat{\Theta}_r^{(3)} = F(Q_x, Q_y, S)$. By the multivariate central limit theorem and Lemma 4.2,

$$(Q_x, Q_y, S) = (1, 1, \rho) + r^{-1/2}\xi + O_{\mathbb{P}}(r^{-1})$$

with ξ asymptotically Gaussian and covariance determined by the moments of (X^2, Y^2, XY). Taylor expansion gives

$$\mathbb{E}F(Q_x, Q_y, S) = F(1, 1, \rho) + \frac{1}{2r}\operatorname{tr}\left(\nabla^2 F(1, 1, \rho)\,\Sigma_\rho\right) + O(r^{-2}),$$

which is (20). The $O_{\mathbb{P}}(r^{-1/2})$ fluctuation is the first-order delta-method term. On the diagonal $X = Y$, ReLU scale invariance makes the normalized correlation and derivative atoms deterministic along the radial direction, cancelling the first mean correction. $\square$

6 Operator Concentration and Spectral Preservation

Let $\Theta_{\infty,L}$ be the deterministic limiting NTK and $\widehat{\Theta}_L$ the empirical finite NTK. Entrywise tensor bounds give

$$\operatorname{Var}((\widehat{\Theta}_L)_{ab}) \lesssim L^3\bar{\varepsilon}, \qquad |\mathbb{E}(\widehat{\Theta}_L)_{ab} - (\Theta_{\infty,L})_{ab}| \lesssim L^3\bar{\varepsilon}. \tag{21}$$

A layer martingale and matrix Freedman yield the operator form. The well-spread data assumption below means that no small group of examples dominates the random feature directions; equivalently, the matrix variance of the empirical NTK fluctuation grows like m, not m^2.

Theorem 6.1 (Finite-network operator concentration)**.** *Assume $L^3\bar{\varepsilon} \le c$. Assume further that the data are well-spread in the layerwise feature bases, so the entrywise tensor variance bounds upgrade to a matrix variance proxy of order $mL^3\bar{\varepsilon}$. With probability at least $1 - \delta$,*

$$\|\widehat{\Theta}_L - \Theta_{\infty,L}\|_{\mathrm{op}} \le CL^{3/2}\sqrt{m\bar{\varepsilon}}\ \mathrm{polylog}(m, L, 1/\delta) + CmL^3\bar{\varepsilon}\ \mathrm{polylog}(m, L, 1/\delta). \tag{22}$$

A robust version replaces the first term by $CmL^{3/2}\sqrt{\bar{\varepsilon}}$ without the well-spread data assumption.

Proof sketch. Expose the random weights and projectors layer by layer and decompose $\widehat{\Theta}_L - \mathbb{E}\widehat{\Theta}_L$ as a matrix martingale. Proposition 5.2 gives conditional entrywise variance $O(L^3\bar{\varepsilon})$ and conditional bias $O(L^3\bar{\varepsilon})$. Under the well-spread data assumption, the predictable quadratic variation is $O(mL^3\bar{\varepsilon})$ in operator norm rather than $O(m^2L^3\bar{\varepsilon})$. Matrix Freedman gives the first term in (22), with logarithmic losses from truncation and union over tensor coordinates. The deterministic bias contributes the second term. Without this assumption, the same argument uses the Frobenius-to-operator bound on the variance proxy and loses a factor $\sqrt{m}$. $\square$

If $\mu_\infty(L) = \lambda_{\min}(P_m\Theta_{\infty,L}P_m)$ is the limiting centered spectral margin, Weyl's inequality gives preservation whenever the right-hand side of (22) is at most $\mu_\infty(L)/2$. In the RF-LR bottleneck, the baseline is not the dense MLP margin; the centered proxy margin can scale as $1/(rL)$. Combining this with the bottleneck deviation gives the conservative design rule $r \gtrsim mL^3$, up to logarithmic factors, when the RF-LR perturbation scale is $1/r$.

6.1 RF-LR memory and rank-depth rules

The RF-LR recursion (8) is not a full-rank-compatible perturbation. Its old-kernel memory obeys an exact product rule:

$$\Theta^{(L)}_{\mathrm{old}\leftarrow k} = \Theta^{(k)} \prod_{j=k+1}^{L} \frac{\dot{\Sigma}^{(j)}}{r}. \tag{23}$$

At the normalized ReLU edge of chaos, $\dot{\Sigma}^{(j)} = 1/2 + O(j^{-1})$, hence

$$\left|\Theta^{(L)}_{\mathrm{old}\leftarrow k}\right| \le C \left(\frac{1}{2r}\right)^{L-k} \left(\frac{L}{k}\right)^{C}. \tag{24}$$

Low rank therefore shortens old NTK memory geometrically. The cost is that the centered deterministic margin can shrink with r and L.

Proposition 6.2 (Unprojected and centered RF-LR stability rules)**.** *Consider an RF-LR bottleneck whose centered limiting margin satisfies $\mu_{\mathrm{RF}}(L,r) \gtrsim (rL)^{-1}$. If the empirical operator deviation is controlled by the conservative scale $L^{3/2}\sqrt{m/r} + mL^3/r$, then spectral preservation follows from*

$$r \gtrsim mL^3 \ \mathrm{polylog}(m, L, 1/\delta). \tag{25}$$

If the radial mode is projected out and an angularized centered decomposition improves the variance scale by one power of L, then the sufficient rule improves to

$$r \gtrsim mL^2 \ \mathrm{polylog}(m, L, 1/\delta). \tag{26}$$

Proof. Weyl's inequality requires the operator deviation to be at most a fixed fraction of $\mu_{\mathrm{RF}}(L,r)$. In the unprojected RF-LR model, the radial contribution remains present in the A-sector, whose tensor scale is cubic in depth. The sufficient inequality is therefore dominated by the mL^3/r contribution after absorbing logarithms and constants, giving (25). In the centered/angularized model, the radial sector is removed before concentration and the dominant tensor variance is quadratic in depth. Repeating the same Weyl argument gives (26). The second statement is conditional on this projection and should not be read as the default RF-LR rule. □

7 NTK Descendants and Training-Time Drift

Let $J_c = \nabla_\theta f(x_c)$ and define the dNTK and ddNTK descendants

$$\Xi_{ab;c} = D\Theta_{ab}[J_c], \qquad \Psi_{ab;cd} = D^2\Theta_{ab}[J_c, J_d]. \tag{27}$$

These descendants add derivative legs to the same layer maps. They do not introduce new independent random variables.

Proposition 7.1 (No new low-rank vertices)**.** *The joint cumulants of (K, Θ, Ξ, Ψ) are generated by the same Gaussian-ReLU atoms and Haar-projector cumulants of $G = (n/r)UU^\top$. Differentiation changes deterministic contractions but not the probabilistic vertex set.*

Proof. Each derivative of the network output with respect to a trainable right factor $B^{(\ell)}$ inserts an additional deterministic backpropagated leg and a factor of the same frozen projector $UU^\top$. Higher derivatives differentiate ReLU gates and covariance maps, changing the local deterministic tensors $\kappa, \dot{\kappa}, \ddot{\kappa}$, but all randomness still comes from the original Gaussian preactivations, readout weights, and frozen projectors. Since U is fixed, there is no moving-projector derivative. Thus the connected diagrams for K, Θ, Ξ, Ψ have the same random vertices as the NTK diagrams and only differ in their deterministic leg labels. □

Under square-loss gradient flow, $\dot{\Theta}_{ab,t} = -\sum_c \Xi_{ab;c,t} e_{c,t}$. The descendant concentration bound therefore yields, in a lazy window,

$$\|\Theta_t - \Theta_0\|_{\mathrm{op}} \le Ct\|e_0\|_2 \left(L^3\sqrt{m\bar{\varepsilon}} + mL^3\bar{\varepsilon}\right) \mathrm{polylog}(m, L, 1/\delta). \tag{28}$$

This closes the O_s hierarchy of Dyer and Gur-Ari [11] under the low-rank rules. If (λ_i, e_i) are eigenpairs of the infinite-width training NTK and Δf_0 is the initial residual, their late-time first NTK correction at a test point can be written

$$\begin{aligned}\Theta^{(1)}_\infty(x) = &-\sum_i \frac{(O_3(x;0)\cdot e_i)(\Delta f_0 \cdot e_i)}{\lambda_i} \\ &+\sum_{i,j} \frac{(e_i^\top O_4(x;0)e_j)(\Delta f_0 \cdot e_i)(\Delta f_0 \cdot e_j)}{\lambda_i(\lambda_i+\lambda_j)}.\end{aligned} \tag{29}$$

Here O_3 and O_4 are contractions of the dNTK and ddNTK descendants. Proposition 7.1 implies that equation (29) remains valid for frozen-left low rank after replacing the dense descendant coefficients by their rank-state/projector diagrams. The small eigenvalue denominators also explain why eigenvector mixing in a power-law tail is the correct observable for the first dynamical correction.

8 Power-Law Data, Spectral Mixing, and Early Stopping

Let the population covariance be

$$\Lambda = \mathrm{diag}(\eta_1, \dots, \eta_N), \qquad \eta_j = \eta_1 j^{-(1+\alpha)}, \quad \alpha > 0, \tag{30}$$

and write the finite NTK training operator as $\widehat{K} = \Lambda + E$, with $\|E\|_{\mathrm{op}} \le s$. Theorem 6.1 supplies the diagrammatic choice

$$s = s_{L,n,r} := C\left[L^{3/2}\sqrt{m\bar{\varepsilon}} + mL^3\bar{\varepsilon}\right] \mathrm{polylog}(m, L, 1/\delta). \tag{31}$$

The adjacent population gap is

$$g_j := \eta_j - \eta_{j+1} = (1+\alpha)\eta_1 j^{-(2+\alpha)}(1 + O(j^{-1})). \tag{32}$$

Theorem 8.1 (Stable head and mixing threshold)**.** *Let q_j be the eigenvector of $\widehat{K}$ associated by eigenvalue order with the population eigenvector e_j, and set $g_0 = +\infty$. For $1 \le j < N$, whenever $\delta_j := \min(g_{j-1}, g_j) > 2s$,*

$$|\widehat{\eta}_j - \eta_j| \le s, \qquad \sin\angle(q_j, e_j) \le \frac{s}{\delta_j - s}. \tag{33}$$

Consequently the number of modes protected uniformly up to constants is

$$J_{\mathrm{mix}} \asymp \left(\frac{(1+\alpha)\eta_1}{2s}\right)^{1/(2+\alpha)}. \tag{34}$$

The bound makes no distributional claim about modes $j \gtrsim J_{\mathrm{mix}}$; it only identifies the tail in which Wigner-like mixing is no longer forbidden by the gaps.

Proof. The eigenvalue statement is Weyl's inequality. The eigenvector statement is the one-dimensional Davis–Kahan theorem, using the perturbed separation at least $\delta_j - s$. Substitution of (32) into $g_j \simeq 2s$ gives (34). □

We next compute the risk if this unresolved tail is orthogonally invariant. For sample size P, inverse noise temperature β, time t, and training eigenvalue $\kappa \ge 0$, define

$$c_t(\kappa) = \frac{1}{2}e^{-2Pt\kappa} + \frac{1 - e^{-2Pt\kappa}}{2P\beta\kappa}, \tag{35}$$

with its continuous extension at $\kappa = 0$. The first term is teacher-averaged bias and the second is the Langevin variance in the power-law regression model of Kramp, Lindner, and Helias [15]. If $\widehat{K} = Q\,\mathrm{diag}(\kappa_i)Q^\top$, the exact linearized risk is

$$\mathcal{L}(t; Q) = \mathrm{tr}\left[\Lambda Q\,\mathrm{diag}(c_t(\kappa_i))Q^\top\right]. \tag{36}$$

Theorem 8.2 (Exact orthogonal-Weingarten tail risk)**.** *Fix a head size $0 \le J \le N-2$. Let $Q = I_J \oplus Q_T$, where Q_T is Haar on $O(M)$, $M = N-J$, and condition on the ordered training eigenvalues κ_i. Write*

$$
\begin{aligned}
S_\eta &= \sum_{i>J} \eta_i, & S_c &= \sum_{i>J} c_t(\kappa_i), \\
S_{\eta,0}^2 &= \sum_{i>J} \eta_i^2 - \frac{S_\eta^2}{M}, & S_{c,0}^2 &= \sum_{i>J} c_t(\kappa_i)^2 - \frac{S_c^2}{M}.
\end{aligned} \tag{37}
$$

Then

$$
\mathbb{E}_{Q_T} \mathcal{L}(t;Q) = \sum_{i \le J} \eta_i c_t(\kappa_i) + \frac{S_\eta S_c}{M}, \tag{38}
$$

$$
\mathrm{Var}_{Q_T} \mathcal{L}(t;Q) = \frac{2 S_{\eta,0}^2 S_{c,0}^2}{(M-1)(M+2)}. \tag{39}
$$

These formulas are finite-N and exact.

Proof. Apply the first and fourth orthogonal Haar moments to $\mathrm{tr}(AQ_T B Q_T^\top)$, with $A = \mathrm{diag}(\eta_{J+1:N})$ and $B = \mathrm{diag}(c_t(\kappa_{J+1:N}))$. The complete invariant calculation is given in Appendix D. □

Corollary 8.3 (Why a fully Haar NTK is the wrong model)**.** *If $J = 0$, the mean risk is $N^{-1}(\sum_i \eta_i)(\sum_i c_t(\kappa_i))$; the mode-by-mode pairing of data eigenvalues and learning rates is lost. Thus a fully Haar eigenbasis generally destroys the aligned power-law scaling. Equations (33)–(39) justify Haar randomization only inside the unresolved tail.*

The aligned dynamics learns modes satisfying $P\eta_j t \gtrsim 1$, so its moving cutoff is $J(t) \asymp (P\eta_1 t)^{1/(1+\alpha)}$. Equating it with (34) gives

$$
t_{\mathrm{mix}} \asymp \frac{1}{P\eta_1} \left(\frac{(1+\alpha)\eta_1}{2s} \right)^{(1+\alpha)/(2+\alpha)}. \tag{40}
$$

Before this time the learned head is perturbatively aligned; after it, training reaches modes for which eigenvectors may mix.

Corollary 8.4 (Rank sufficient to protect early stopping)**.** *The aligned power-law optimum is*

$$
t_0^* = \frac{\beta}{2} \frac{\alpha}{1+\alpha} \tag{41}
$$

to leading order [15]. A sufficient condition for $t_0^* \lesssim t_{\mathrm{mix}}$ is

$$
\frac{s}{\eta_1} \lesssim C_\alpha (P\beta\eta_1)^{-(2+\alpha)/(1+\alpha)}. \tag{42}
$$

In the low-rank dominated regime $\bar{\varepsilon} \asymp r^{-1}$, the variance term of (31) therefore yields

$$
r \gtrsim mL^3 (P\beta\eta_1)^{2(2+\alpha)/(1+\alpha)} \,\mathrm{polylog}(m, L, P, 1/\delta). \tag{43}
$$

The bias term gives the weaker power $r \gtrsim mL^3(P\beta\eta_1)^{(2+\alpha)/(1+\alpha)}$ when $P\beta\eta_1 \gtrsim 1$.

Proof. Insert (41) into $t_0^* \lesssim t_{\mathrm{mix}}$, rearrange, and then substitute $s \asymp L^{3/2}\sqrt{m/r}$ or $s \asymp mL^3/r$ from (31). □

9 Experiments

We run four classes of experiments and present the figures in the same order as the theory. Proxy experiments first validate the rank-projector law and operator normalization. We then check the endpoint tensor constants and RF-LR memory suppression. True finite-network experiments compute empirical NTKs by autodiff and estimate direct finite-network covariances of the NNGP Gram and

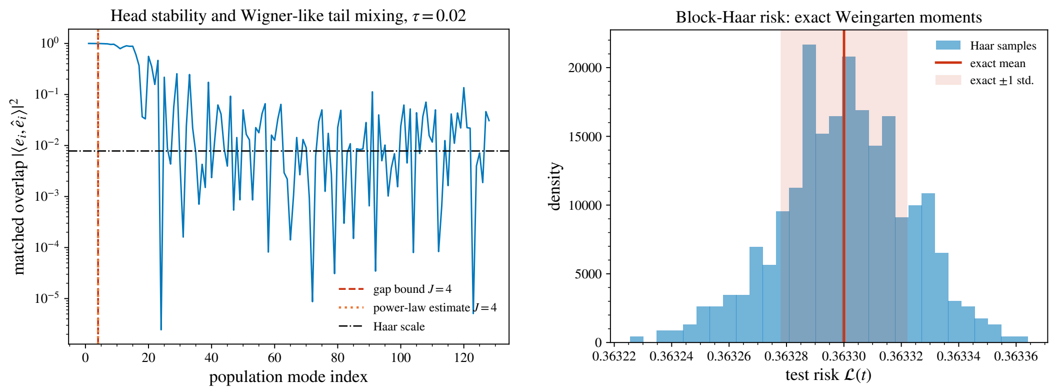

Figure 3: Spectral mixing and exact orthogonal moments. Left: population-eigenvector overlap under the PSD Wigner deformation; the exact gap threshold and the asymptotic power-law estimate coincide at $J_{\text{mix}} = 4$. Right: the block-Haar test-risk distribution with the exact finite-dimensional Weingarten mean and one-standard-deviation band.

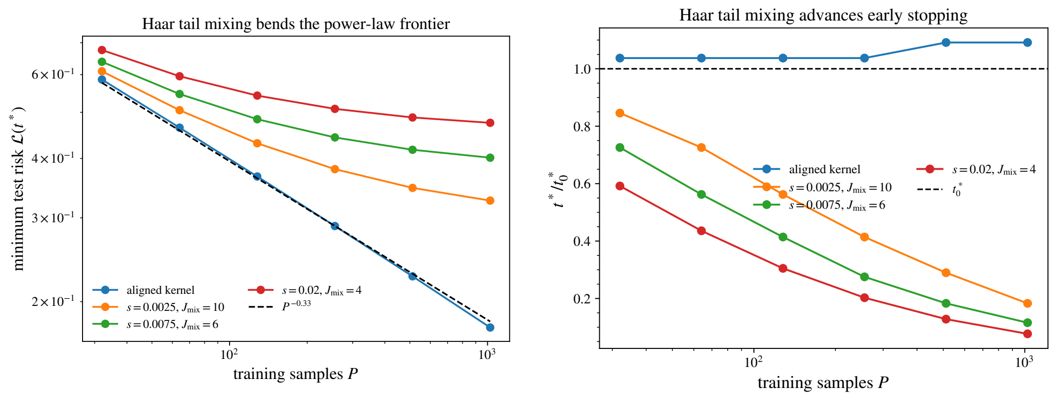

Figure 4: Consequences of a Haar-mixed power-law tail. Left: the aligned $P^{-1/3}$ risk frontier persists until the training cutoff reaches the tail, after which it bends toward a perturbation-controlled plateau. Right: tail mixing advances the optimal stopping time, increasingly so as P resolves deeper modes.

empirical NTK. Finally, power-law experiments test the exact Weingarten moments, the stable-head threshold, and the predicted early-stopping failure mode.

For the last group we use $\alpha = 1/2$, $\eta_1 = \beta = 1$, and $N = 128$ modes for the Wigner deformation

$$\widehat{K} = (\Lambda^{1/2} + \tau W)(\Lambda^{1/2} + \tau W)^\top, \qquad W \text{ GOE-normalized}, \tag{44}$$

which is positive semidefinite while having a Wigner-like leading correction. At $\tau = 0.02$, the measured perturbation norm is $s = 0.02178$; both the exact gap count and equation (34) give $J_{\text{mix}} = 4$. Figure 3 shows that these first modes retain large population overlap while the tail fluctuates near the Haar scale. In a separate block-Haar test with head size 32, 500 orthogonal samples give empirical mean 0.36330037 and standard deviation 2.249×10^{-5}, versus the exact values 0.36330004 and 2.187×10^{-5} from Theorem 8.2. The mean discrepancy is 0.34 Monte–Carlo standard errors.

To isolate data scaling, we use $N = 8192$ exact power-law modes and replace only the tail beyond J_{mix} by a Haar block. The aligned minimum follows $P^{-\alpha/(1+\alpha)} = P^{-1/3}$ and its numerical optimum $t^* = 0.173$–0.182 is close to $t_0^* = 1/6$. Once the learned cutoff enters the mixed tail, the minimum-risk curve bends above that frontier and the optimum moves earlier. For example, at $P = 1024$, perturbation scales $s = 0.0025, 0.0075, 0.02$ give $J_{\text{mix}} = 10, 6, 4$ and optimal-time ratios $t^*/t_0^* = 0.183, 0.116, 0.077$, respectively. Figure 4 is an exact evaluation of equations (38)–(39), not a fitted surrogate.

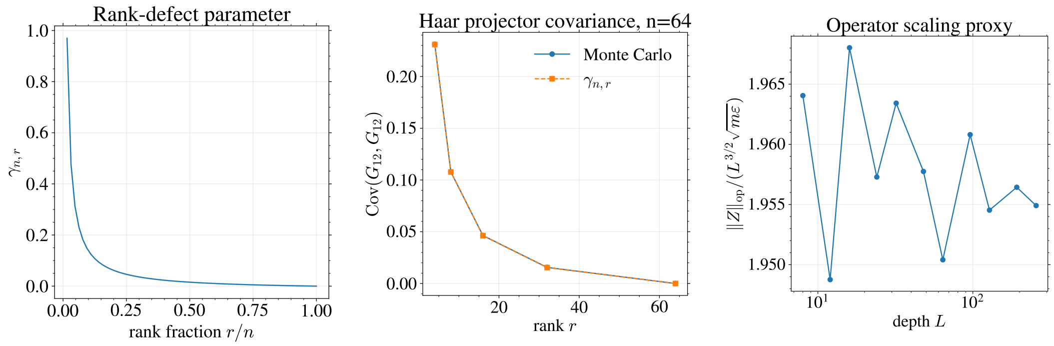

Figure 5: Rank-projector and operator checks. Left: the rank-defect scale $\gamma_{n,r}$ vanishes at full rank. Middle: empirical Haar-projector covariance matches the closed form. Right: the normalized operator proxy $\|Z\|_{\mathrm{op}}/(L^{3/2}\sqrt{m\varepsilon})$ is approximately stable across depth.

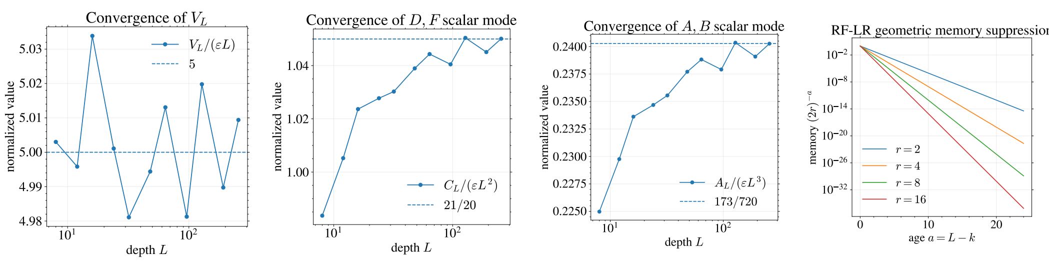

Figure 6: Tensor constants and RF-LR memory. The first three panels show convergence to the isolated constants 5, $21/20$, and $173/720$. The last panel shows geometric suppression of old NTK memory in the RF-LR bottleneck.

10 Conclusion

Low rank changes finite-width NTK theory not only by reducing parameters, but by adding a second family of connected Feynman vertices. For frozen-left/trainable-right matrices, these vertices are exactly the cumulants of a random projector; their second-order weight is $\gamma_{n,r}$, and they disappear pathwise at full rank. The external transport then fixes the universal critical-depth degrees $1, 2, 3$ for V_4, (D, F), and (A, B). This gives a diagrammatic explanation of the effective slopes in the published dense-network plots and a full-rank-compatible expansion for concatenated MMNN blocks.

The operator consequence is spectral rather than merely entrywise. In a power-law data model, finite-rank corrections preserve a Davis–Kahan head but may randomize the near-degenerate tail. Orthogonal Weingarten calculus computes the entire block-Haar tail risk exactly and turns the diagrammatic perturbation scale into a mixing time and an early-stopping rank law. The theory therefore predicts two distinct transitions: the familiar cubic depth-rank threshold $r \gtrsim mL^3$ for an NTK convexity certificate, and the stronger data-dependent condition (43) required to retain aligned neural scaling laws at the noise-optimal stopping time.

The scope is explicit. The exact results cover signed/isometric frozen-left factors and the stated block-Haar tail model; strict positive NMF has cone-projected gradients and requires additional boundary vertices. Haar behavior is neither assumed nor claimed for the protected spectral head. The tensor hierarchy and operator concentration retain their stated transport and well-spread-data hypotheses, and the constants 5, $21/20$, and $173/720$ are isolated pairing-basis coefficients rather than arbitrary scalar-output cumulants.

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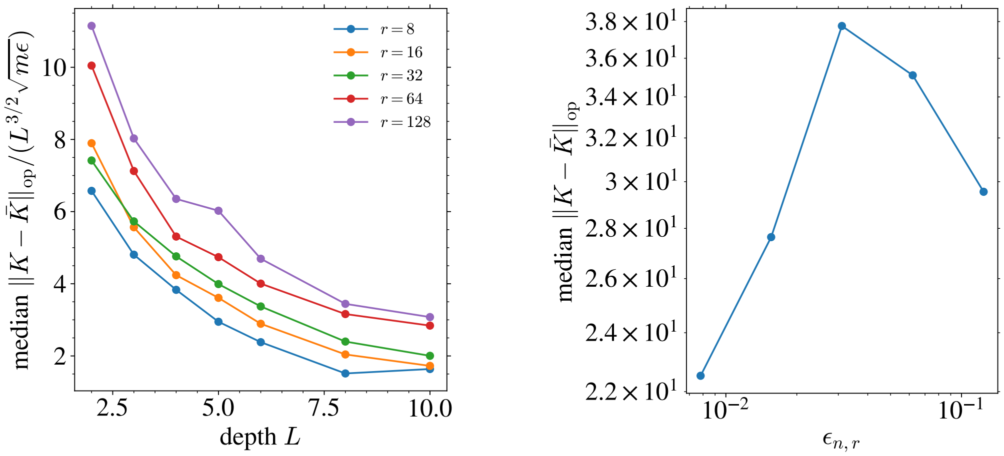

Figure 7: True finite-network NTK checks computed by autodiff. Left: empirical operator deviations follow the predicted normalization. Right: the finite-network deviation tracks the rank-defect scale and vanishes at the full-rank endpoint.

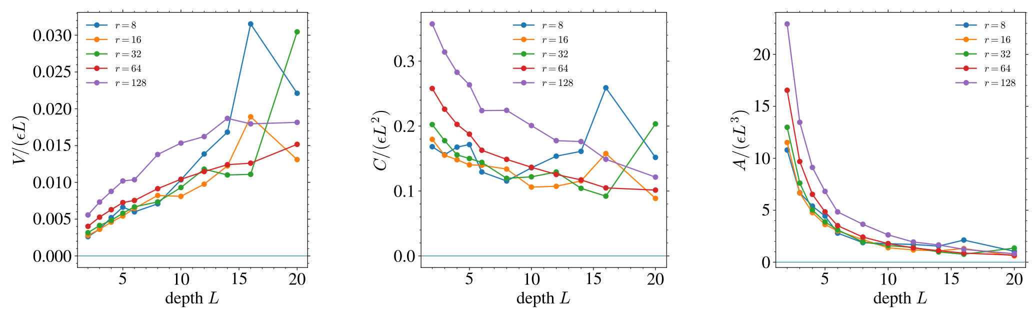

Figure 8: True finite-network cumulants from autodiff NTKs. These scalar-output cumulants are not isolated Feynman coefficients; they test the predicted polynomial control and rank-width scaling on actual finite networks.

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Table 2: Exact full-rank matching for true empirical NTKs at $n = r = 128$, $m = 8$, $d = 16$. Differences remain at numerical precision up to depth $L = 200$, confirming Theorem 3.1.

Depth	max absolute NTK difference	relative difference
2	$8.88 \cdot 10^{-16}$	$8.09 \cdot 10^{-17}$
4	$3.55 \cdot 10^{-15}$	$1.31 \cdot 10^{-16}$
6	$5.33 \cdot 10^{-15}$	$4.05 \cdot 10^{-16}$
8	$3.55 \cdot 10^{-15}$	$4.20 \cdot 10^{-16}$
10	$6.22 \cdot 10^{-15}$	$1.14 \cdot 10^{-15}$
20	$1.95 \cdot 10^{-14}$	$3.77 \cdot 10^{-15}$
50	$5.68 \cdot 10^{-14}$	$6.68 \cdot 10^{-16}$
100	$2.14 \cdot 10^{-15}$	$2.35 \cdot 10^{-14}$
200	$8.08 \cdot 10^{-14}$	$1.27 \cdot 10^{-14}$

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A Full Proof of the Frozen-Left Decomposition

We prove Proposition 3.2. Consider one layer

$$W = \alpha U B, \qquad U^\top U = I_r, \qquad B \in \mathbb{R}^{r \times d},$$

and let $H = \partial \mathcal{L}/\partial W$ be the Euclidean loss gradient in dense-weight coordinates. The Frobenius inner product is used throughout.

First freeze U. For a variation dB,

$$dW = \alpha U\, dB, \qquad d\mathcal{L} = \langle H, \alpha U\, dB \rangle = \langle \alpha U^\top H, dB \rangle.$$

Hence $\nabla_B \mathcal{L} = \alpha U^\top H$. Gradient descent in B gives

$$\dot{B} = -\eta\alpha U^\top H, \qquad \dot{W}_B = \alpha U\dot{B} = -\eta\alpha^2 UU^\top H,$$

which is (6). Therefore the B-only tangent directions are exactly the fixed linear subspace

$$\mathcal{T}_B W = \{\alpha U\Delta B : \Delta B \in \mathbb{R}^{r\times d}\}.$$

The corresponding contribution to the empirical NTK is a fixed-projector block, because every dense gradient is projected through $UU^\top$.

Now allow U to vary on the Stiefel manifold $\mathrm{St}(n,r) = \{U : U^\top U = I_r\}$. Its tangent space is

$$T_U \mathrm{St}(n,r) = \{\Delta U : U^\top \Delta U + \Delta U^\top U = 0\}.$$

Every tangent vector decomposes as

$$\Delta U = U\Omega + U_\perp K, \qquad \Omega^\top = -\Omega, \qquad U^\top U_\perp = 0,$$

where $U\Omega$ rotates the basis inside $\mathrm{span}(U)$ and $U_\perp K = (I - UU^\top)\Delta U$ moves the subspace itself. For a variation dU,

$$dW = \alpha\, dU\, B, \qquad d\mathcal{L} = \langle H, \alpha dUB\rangle = \langle \alpha HB^\top, dU\rangle.$$

Thus the Euclidean gradient in U-coordinates is $\alpha HB^\top$. Its normal subspace-moving component is

$$(I - UU^\top)\alpha HB^\top.$$

The skew component $U\Omega$ only rotates coordinates inside the same represented subspace: by replacing U with UR and B with $R^\top B$, the product UB is unchanged. This is why the main text states (7) up to rotations inside $\mathrm{span}(U)$ that can be absorbed into B.

Keeping only the subspace-moving component, Stiefel gradient descent contributes

$$\dot{U}_\perp = -\eta\alpha(I - UU^\top)HB^\top.$$

Mapping this velocity back to dense-weight coordinates gives

$$\dot{W}_U = \alpha\dot{U}_\perp B = -\eta\alpha^2(I - UU^\top)HB^\top B,$$

which is (7). This term is absent when U is frozen. It depends on the current right factor through $B^\top B$, so it cannot be represented by cumulants of the fixed Haar projector $G = (n/r)UU^\top$ alone.

Finally, if both U and B are trained, the tangent space contains the sum of B-directions $\alpha U\Delta B$, Stiefel subspace directions $\alpha\Delta UB$, and their cross terms in the NTK Gram matrix. Along gradient flow these directions also vary in time, producing dNTK terms involving $\dot{U}$, $\dot{B}$, and products such as $B^\top B$. The frozen-left model deliberately removes these moving-subspace interactions, leaving a closed fixed-projector diagrammatic calculus.

B Projector Cumulants Beyond Second Order

For $G = (n/r)UU^\top$, the mean is $\mathbb{E}G = I_n$, the trace is deterministic, and every connected cumulant containing the trace direction vanishes. Orthogonal Weingarten calculus gives, for fixed cumulant order s,

$$\mathrm{cum}(G_{i_1 j_1}, \ldots, G_{i_s j_s}) = O_s(r^{1-s}) \tag{45}$$

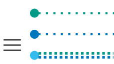
uniformly when $r \le n$ and $r \to \infty$, with the full-rank endpoint cancelling after replacing r^{-1} by the defect scale $\gamma_{n,r}$ at second order. Diagrammatically, this is why a connected rank vertex is counted by the number of connected projector insertions, not simply by the number of factors U in the algebraic expression.

C Low-Rank Diagrammatic Rules

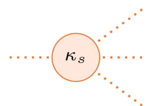
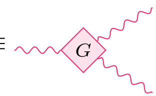
This appendix records the graphical calculus underlying the perturbative statements in the main text. The external objects are the same as in finite-width NTK Feynman diagrams: NNGP/preactivation legs, NTK fluctuation legs, and derivative-NTK descendants. The difference is internal. In the low-rank model, stochastic vertices are empirical rank cumulants and frozen-projector cumulants rather than unconstrained neural-index contractions.

C.1 Visual dictionary

We use the same visual grammar as the finite-width NTK diagrammatics: filled external dots denote observed sample indices, solid lines denote NNGP/preactivation objects, colored dotted ghost lines denote NTK fluctuations, and multi-line ghost objects denote dNTK/ddNTK descendants. Low-rank only changes the internal stochastic vertices.

$z_\alpha \equiv$ $\qquad \widehat{\Delta\Theta}_{\alpha\beta} \equiv$  $\qquad \widehat{\mathrm{d}\Theta} \equiv$  . (46)

The new internal low-rank vertices are:

rank cumulant $\equiv$  $\sim r^{1-s}\kappa_s$, $\qquad$ projector cumulant $\equiv$  $\sim \mathrm{cum}(G, \ldots, G)$. (47)

C.2 Rank-state vertices

Let

$$M_A^{(\ell)} = \frac{1}{r_\ell}\sum_{a=1}^{r_\ell} \phi_A^{(\ell)}(Z_a), \tag{48}$$

where A labels a local atom such as $\sigma\sigma$, $\sigma\sigma'$, $\sigma'\sigma'$, or a derivative analogue. Conditional on the past, the rank atoms are iid. Therefore Z_a is the same channel-local random object defined in Lemma 4.2, with the layer superscript suppressed.

$$\mathrm{cum}(M_{A_1}^{(\ell)}, \ldots, M_{A_s}^{(\ell)}) = r_\ell^{1-s}\kappa_{A_1\cdots A_s}^{(\ell)}, \qquad \kappa_{A_1\cdots A_s}^{(\ell)} = \mathrm{cum}(\phi_{A_1}^{(\ell)}(Z), \ldots, \phi_{A_s}^{(\ell)}(Z)). \tag{49}$$

Thus every connected rank vertex of valence s has weight r_ℓ^{1-s}. In the isometric model, one also has frozen-projector vertices given by cumulants of $G = (n/r)UU^\top$. At valence two this is Lemma 4.1; at higher valence the weights are orthogonal Weingarten contractions. When $r = n$, $G = I$, so every centered projector vertex vanishes.

Theorem C.1 (Low-rank Feynman rules)**.** *Let F be a smooth observable of finitely many layer states $M^{(\ell)}$ and frozen projectors $G^{(\ell)}$. Its expansion in powers of $1/r_\ell$ and projector cumulants is obtained by summing connected low-rank diagrams with the prescribed external legs. Algebraically:*

1. *a rank vertex of valence s contributes $r_\ell^{1-s}\kappa_s^{(\ell)}$;*
2. *a projector vertex contributes the corresponding cumulant of $G^{(\ell)}$, with second-order scale γ_{n_ℓ, r_ℓ};*
3. *a deterministic layer vertex contributes a Frechet derivative $D^q\Phi_\ell$ of the layer map;*
4. *a transport edge contributes the product of first derivatives along the depth path;*
5. *for cumulants, disconnected diagrams cancel by the moment-cumulant relation.*

Proof. Taylor expand F around the deterministic trajectory of the rank states. The cumulant generating operator for an empirical rank average is

$$\exp\left(\sum_{s\geq 2} \frac{r^{1-s}}{s!}\kappa_{A_1\cdots A_s}\partial_{A_1}\cdots\partial_{A_s}\right), \tag{50}$$

up to the usual finite-order smoothness remainder. Expanding (50) gives the rank vertices and their deterministic derivative contractions. The same argument applies to G, replacing empirical-rank cumulants by frozen-projector cumulants. Taking connected cumulants removes disconnected products. ☐

Proposition C.2 (Combined full-width/rank power counting)**.** *Consider a correlation function with m_Γ derivative-tensor vertices and Dyer–Gur-Ari cluster graph having n_e even components and n_o odd components. Assume its dense channel diagrams obey the cluster-graph bound*

$$s_C = n_e + \frac{n_o}{2} - \frac{m_\Gamma}{2}. \tag{51}$$

For a low-rank diagram Γ, let $\mathcal{R}(\Gamma)$ be its empirical-rank vertices, with valence d_v, and $\mathcal{P}(\Gamma)$ its centered projector vertices. At fixed depth,

$$|\mathcal{A}_\Gamma| \le C_\Gamma n^{s_C} \prod_{v\in\mathcal{R}(\Gamma)} r_{\ell(v)}^{1-d_v} \prod_{v\in\mathcal{P}(\Gamma)} \left\| \mathrm{cum}_{d_v}(G^{(\ell(v))}) \right\|, \tag{52}$$

times the deterministic depth transports of Theorem C.1. A second projector vertex contributes γ_{n_ℓ,r_ℓ}; more generally a connected valence-d rank source is $O(r_\ell^{1-d})$. At $r_\ell = n_\ell$, every centered projector diagram is zero and the surviving diagrams are exactly the dense ones.

Proof. Before the low-rank averages are taken, the dense Gaussian part has the same neural-index loops and the same $n^{-m_\Gamma/2}$ normalization as in the cluster-graph expansion. The number of loops is at most $n_e + n_o/2$, giving n^{s_C}. Conditional independence across rank channels forces all legs at a connected empirical-rank vertex to share one channel index; summing it gives r, while its d_v normalized legs give r^{-d_v}. Frozen-projector blocks are independent across layers and contribute their connected cumulants. Multiplication yields (52); deterministic propagation does not change width/rank counting. Finally $G = I$ at $r = n$, so all its centered cumulants vanish, while Theorem 3.1 identifies the remaining diagrams with the dense expansion. □

C.3 Basic tensor recursions

The diagrammatic rules produce the same external tensor hierarchy as the dense finite-width expansion, but with low-rank sources. Let J_ℓ^K and J_ℓ^Θ denote the deterministic linear transports for NNGP and NTK legs. The NNGP covariance block satisfies schematically

$$V_{\ell+1} = J_\ell^K V_\ell (J_\ell^K)^\top + \Xi_\ell^{KK}, \qquad \Xi_\ell^{KK} = r_\ell^{-1}\kappa_2(S^K, S^K) + \gamma_{n_\ell,r_\ell}\mathfrak{h}_\ell^{KK}. \tag{53}$$

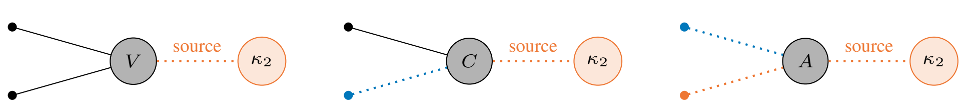
For the mixed NNGP-NTK block $C = (D, F)$,

$$C_{\ell+1} = J_\ell^K C_\ell (J_\ell^\Theta)^\top + \mathsf{S}_V[V_\ell] + \Xi_\ell^{K\Theta}. \tag{54}$$

For the NTK-NTK block $A = (A, B)$,

$$A_{\ell+1} = J_\ell^\Theta A_\ell (J_\ell^\Theta)^\top + \mathsf{S}_C[C_\ell] + \mathsf{S}_V[V_\ell] + \Xi_\ell^{\Theta\Theta}. \tag{55}$$

The sources Ξ^{KK}, $\Xi^{K\Theta}$, and $\Xi^{\Theta\Theta}$ are precisely the rank and projector cumulant vertices with the corresponding external legs. This explains why the depth exponents in Proposition 5.2 come from external transport, while the small parameters come from r^{-1}, $1/n$, and $\gamma_{n,r}$.



(56)

C.4 First NTK mean correction

The familiar finite-width correction to the NTK mean has the same external five-term form:

$$\Theta_{\ell+1}^{\{1\}} = \mathsf{T}_\Theta[\Theta_\ell^{\{1\}}] + \mathsf{T}_K[K_\ell^{\{1\}}] + \mathsf{S}_V[V_\ell] + \mathsf{S}_D[D_\ell] + \mathsf{S}_F[F_\ell]. \tag{57}$$

The low-rank difference is that the source tensors V, D, F are further resolved into the rank-state and projector vertices above. Thus the low-rank calculus does not change the external NTK grammar; it changes the internal expansion parameter from neural-index contractions to empirical-rank cumulants and frozen-projector cumulants.

C.5 Pairing extraction

A rank-four orthogonally invariant tensor decomposes as

$$C_{i_1 i_2 i_3 i_4} = c_1 \delta_{i_1 i_2} \delta_{i_3 i_4} + c_2 \delta_{i_1 i_3} \delta_{i_2 i_4} + c_3 \delta_{i_1 i_4} \delta_{i_2 i_3}. \tag{58}$$

Let B_1, B_2, B_3 be the three pairings. The normalized pairing Gram matrix is

$$\overline{G}_N = \begin{pmatrix} 1 & 1/N & 1/N \\ 1/N & 1 & 1/N \\ 1/N & 1/N & 1 \end{pmatrix}, \qquad \overline{G}_N^{-1} = \frac{N}{N-1} I_3 - \frac{N}{(N-1)(N+2)} \mathbf{1}\mathbf{1}^\top. \tag{59}$$

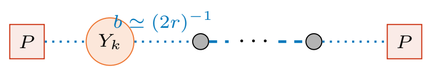
If $h = (\langle C, B_1 \rangle, \langle C, B_2 \rangle, \langle C, B_3 \rangle)$ are measured contractions, the isolated tensor coefficients are $(c_1, c_2, c_3)^\top = \overline{G}_N^{-1} h$. This is why direct scalar-output cumulants should not be identified with the endpoint coefficients 5, $21/20$, and $173/720$ without a pairing-basis extraction.

C.6 Centered RF-LR memory lemma

For RF-LR, an old NTK line transported from layer k to L carries

$$B_{L:k} = \prod_{s=k+1}^{L} \frac{1}{r} \dot{\Sigma}^{(s)}. \tag{60}$$

At ReLU edge of chaos, $\dot{\Sigma}^{(s)} \to 1/2$, so $\|B_{L:k}\| \lesssim (2r)^{-(L-k)}$, up to polynomial endpoint factors. The corresponding centered transport diagram is



(61)

Let $P = I - \mathbf{1}\mathbf{1}^\top/m$ and $Z_L = P(\widehat{K}_L - K_L)P$. The centered fluctuation admits the linearized expansion

$$Z_L = \sum_{k=1}^{L} B_{L:k+1} Y_k + \sum_{k=1}^{L} B_{L:k+1} R_k, \tag{62}$$

where Y_k is a centered fresh rank source and R_k is quadratic in earlier fluctuations.

Lemma C.3 (Centered RF-LR memory)**.** *Assume, conditionally on the past,*

$$\left\| \mathbb{E}[Y_k^2 \mid \mathcal{F}_{k-1}] \right\|_{\mathrm{op}} \le C \frac{m \varepsilon_{\mathrm{RF}}}{r^2}, \qquad \|Y_k\|_{\psi_1} \le C \frac{\sqrt{\varepsilon_{\mathrm{RF}}}}{r}, \tag{63}$$

and $\|B_{L:k}\| \le q^{L-k}$ *for some* $q < 1$*. If* $\|R_k\|_{\mathrm{op}} \le C m \varepsilon_{\mathrm{RF}}/r^2$ *with high probability, then, up to logarithmic factors,*

$$\|Z_L\|_{\mathrm{op}} \lesssim \frac{\sqrt{m \varepsilon_{\mathrm{RF}} \log(m/\delta)}}{r} + \frac{m \varepsilon_{\mathrm{RF}} \log(m/\delta)}{r^2} \tag{64}$$

with probability at least $1 - \delta$*.*

Proof. Set $X_k = B_{L:k+1} Y_k$. The predictable matrix variance satisfies

$$\left\| \sum_{k=1}^{L} \mathbb{E}[X_k^2 \mid \mathcal{F}_{k-1}] \right\|_{\mathrm{op}} \le C \frac{m \varepsilon_{\mathrm{RF}}}{r^2} \sum_{k=1}^{L} q^{2(L-k)} \le C' \frac{m \varepsilon_{\mathrm{RF}}}{r^2}. \tag{65}$$

Matrix Freedman or Bernstein gives the first term in (64). For the quadratic remainder,

$$\left\| \sum_{k=1}^{L} B_{L:k+1} R_k \right\|_{\mathrm{op}} \le C \frac{m \varepsilon_{\mathrm{RF}}}{r^2} \sum_{k=1}^{L} q^{L-k} \le C' \frac{m \varepsilon_{\mathrm{RF}}}{r^2}. \tag{66}$$

□

Combining Lemma C.3 with Weyl's inequality gives the centered/angularized criterion. If the centered deterministic margin satisfies $\lambda_{\min}(P K_L P) \gtrsim (rL)^{-1}$, and if $\varepsilon_{\mathrm{RF}} \simeq 1/r$, then the fluctuation bound is below the margin when

$$r \gtrsim m L^2 \tag{67}$$

up to logarithms. Without removing the radial constant mode, a diffusive $\sqrt{L}$ factor can reappear, giving the conservative cubic-depth criterion used for the unprojected RF-LR statement.

D Proofs for Power-Law Spectral Mixing

D.1 Aligned learning curve and optimal time

Let $q = 1 + \alpha$, $a = \alpha/(1 + \alpha)$, and $p = 1/(1 + \alpha)$. For an aligned kernel $\widehat{K} = \Lambda$, equation (36) separates as

$$\mathcal{L}_{\text{bias}}(t) = \frac{1}{2}\sum_{j=1}^{N} \eta_j e^{-2Pt\eta_j}, \tag{68}$$

$$\mathcal{L}_{\text{var}}(t) = \frac{1}{2P\beta}\sum_{j=1}^{N}(1 - e^{-2Pt\eta_j}). \tag{69}$$

In the scaling window $1 \ll 2Pt\eta_1 \ll N^{1+\alpha}$, Euler–Maclaurin summation gives

$$\mathcal{L}_{\text{bias}}(t) = \frac{\eta_1 \Gamma(a)}{2(1+\alpha)}(2Pt\eta_1)^{-a}(1 + o(1)), \tag{70}$$

$$\mathcal{L}_{\text{var}}(t) = \frac{\Gamma(a)}{2P\beta}(2Pt\eta_1)^{p}(1 + o(1)). \tag{71}$$

Indeed, with $u = 2Pt\eta_1 x^{-q}$,

$$\int_0^\infty \eta_1 x^{-q} e^{-2Pt\eta_1 x^{-q}}\,\mathrm{d}x = \frac{\eta_1}{q}(2Pt\eta_1)^{-a}\Gamma(a), \tag{72}$$

$$\int_0^\infty (1 - e^{-2Pt\eta_1 x^{-q}})\,\mathrm{d}x = (2Pt\eta_1)^{p}\Gamma(a). \tag{73}$$

Differentiating (70)–(71) yields $2Pt\eta_1 = aP\beta\eta_1$, hence

$$t_0^* = \frac{a\beta}{2} = \frac{\beta}{2}\frac{\alpha}{1+\alpha}, \qquad \mathcal{L}(t_0^*) \asymp \eta_1^{1/(1+\alpha)}(P\beta)^{-\alpha/(1+\alpha)}. \tag{74}$$

This proves the aligned scaling statements used in Section 8; finite-N tail subtraction produces the familiar $O(N^{-\alpha})$ correction.

D.2 Orthogonal fourth moment

We prove Theorem 8.2. The following identity is the finite-dimensional orthogonal Weingarten calculation behind it.

Lemma D.1 (Orthogonal twirl)**.** *Let Q be Haar on $O(M)$, $M \geq 2$, and let A, B be deterministic real symmetric matrices. With $A_0 = A - (\operatorname{tr} A/M)I$ and $B_0 = B - (\operatorname{tr} B/M)I$,*

$$\mathbb{E}\operatorname{tr}(AQBQ^\top) = \frac{\operatorname{tr} A \ \operatorname{tr} B}{M}, \tag{75}$$

$$\operatorname{Var}\operatorname{tr}(AQBQ^\top) = \frac{2\operatorname{tr}(A_0^2)\operatorname{tr}(B_0^2)}{(M-1)(M+2)}. \tag{76}$$

Proof. Haar invariance implies $\mathbb{E}[QBQ^\top] = (\operatorname{tr} B/M)I$, proving the mean. Constant trace components have no variance, so it remains to consider A_0, B_0 in the space S_0^M of traceless symmetric matrices. The orthogonal action $B_0 \mapsto QB_0Q^\top$ is irreducible on this space. Therefore its invariant covariance is

$$\mathbb{E}\big[\langle A_0, QB_0Q^\top\rangle_F^2\big] = c_B\|A_0\|_F^2. \tag{77}$$

Summing this identity over an orthonormal basis of S_0^M gives $\|B_0\|_F^2 = c_B \dim(\mathsf{S}_0^M)$. Since $\dim(\mathsf{S}_0^M) = M(M+1)/2 - 1 = (M-1)(M+2)/2$, equation (76) follows. This is the invariant form of the orthogonal fourth-moment/Weingarten contraction [16]. □

Set $A = \operatorname{diag}(\eta_{J+1:N})$, $B = \operatorname{diag}(c_t(\kappa_{J+1:N}))$, and add the deterministic aligned-head contribution. Then $\operatorname{tr}(A_0^2) = S_{\eta,0}^2$ and $\operatorname{tr}(B_0^2) = S_{c,0}^2$, so Lemma D.1 gives equations (38)–(39).

D.3 Reproducibility

The power-law figures and all plotted CSV values are generated by `experiments/feynman/run_powerlaw_early_stopping.py` with seed 20260806, $N = 128$, $\alpha = 1/2$, $\beta = 1$, $P = 64$, 16 Wigner repetitions, 500 Weingarten repetitions, and an 8192-mode exact block-Haar scaling calculation. The automated tests in `tests/test_powerlaw_weingarten.py` check the aligned risk identity, the orthogonal-twirl degeneracy for a constant spectrum, and the zero-variance block-Haar limit at $t = 0$.

E Why Strict Positive NMF Is Not Covered

The model in (1) is signed and isometric. The right factor B is unconstrained and the frozen left factor U has orthonormal columns. Strict nonnegative matrix factorization imposes a cone constraint and changes the tangent space even at initialization. Its NTK is a projected or conic kernel, not the fixed linear-subspace kernel analyzed here. Therefore the results should not be cited as a theorem for positive NMF without a separate analysis of active constraints, boundary faces, and cone-projected gradients.

F Proof Details for Endpoint Constants

For $g \sim \mathcal{N}(0,1)$, let $X_0 = 2(g_+)^2$ and $Y_0 = 2\mathbf{1}_{\{g>0\}}$. Since $\mathbb{E}[g_+^2] = 1/2$ and $\mathbb{E}[g_+^4] = 3/2$, we have $\mathbb{E}X_0 = 1$, $\mathbb{E}X_0^2 = 6$, and $\mathrm{Var}(X_0) = 5$. Also $\mathbb{E}Y_0 = 1$, $\mathbb{E}Y_0^2 = 2$, so $\mathrm{Var}(Y_0) = 1$. Finally, $\mathbb{E}X_0Y_0 = 2$, hence $\mathrm{Cov}(X_0, Y_0) = 1$. Transporting one or two NTK legs and applying the ReLU edge-of-chaos transport sum gives the formulas for B_λ and $A_{\lambda,\mu}$.

G Additional Experimental Figures

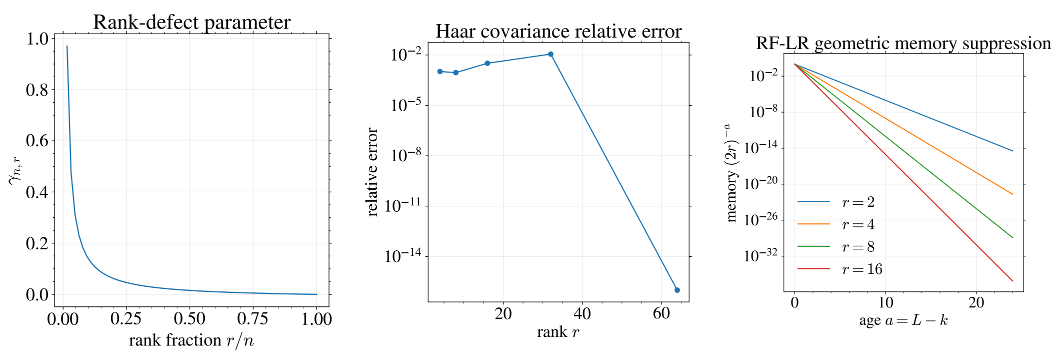


Figure 9: Additional proxy diagnostics: rank-defect parameter, relative Haar covariance error, and RF-LR geometric memory suppression.

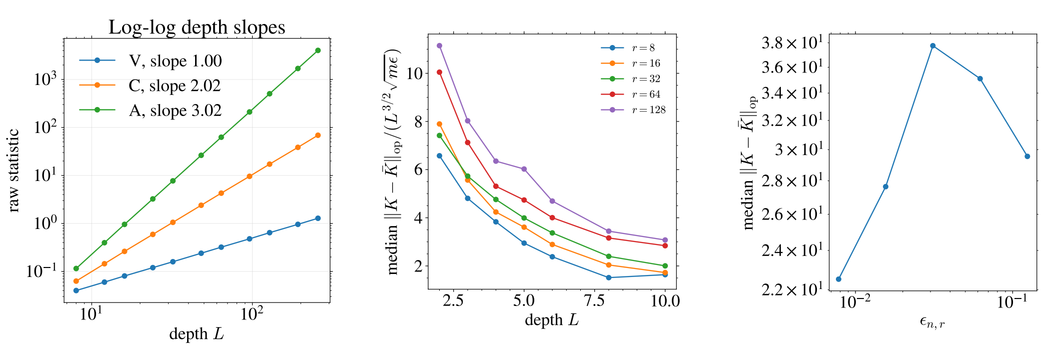


Figure 10: Depth-slope proxy diagnostics and true finite-network NTK operator/rank-defect checks.

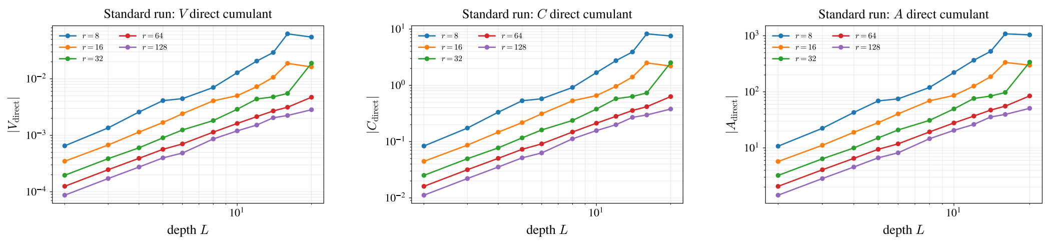


Figure 11: Log-log direct finite-network cumulant magnitudes. These plots complement the normalized-ratio plots in the main text.

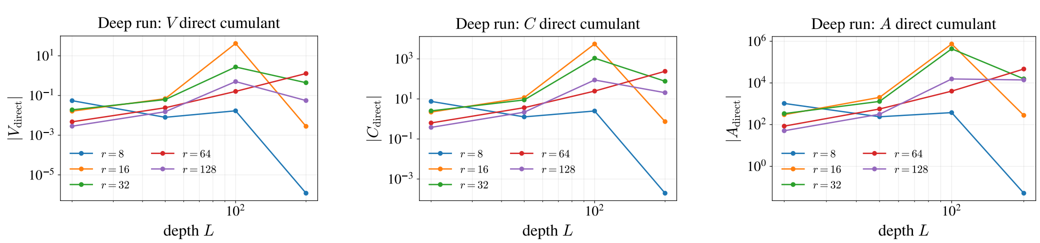


Figure 12: Deep-run log-log cumulant magnitudes for $L \in \{20, 50, 100, 200\}$. These plots are a stress test rather than a clean asymptotic fit: they reveal where finite-network cumulants become dominated by finite-depth and sampling effects.